Included Variable Bias in Claassen (2020): An Empirical Application

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1 Introduction

Claassen (2020, AJPS) estimates the effect of public support for democracy on subsequent democratic change using an autoregressive distributed lag (ADL) model with country-year panel data. The main specification is:

$$d_{it} = \alpha + \phi_1 d_{it-1} + \phi_2 d_{it-2} + \beta s_{it-1} + \mathbf{Z}'_{it-1}\gamma + \varepsilon_{it}$$

where d_{it} is democracy (V-Dem Liberal Democracy index), s_{it-1} is lagged public support for democracy, and $\mathbf{Z}_{it-1}$ is a vector of five controls: log GDP per capita, GDP growth, regional democracy, percent Muslim, and resource dependence.

A key question is whether any of these controls are **colliders** — variables caused by both the treatment (s_{it-1}) and the outcome (d_{it}). If so, conditioning on them introduces **Included Variable Bias** (IVB). The IVB for each candidate collider z is given by:

$$\text{IVB}(\hat{\beta}) = \hat{\beta}^* - \hat{\beta} = -\hat{\theta}^* \cdot \hat{\pi}$$

where $\hat{\theta}^*$ is the coefficient of z in the "long" model (with z) and $\hat{\pi}$ is the coefficient of the treatment in the auxiliary regression of z on the treatment and other controls.

We apply this decomposition to Claassen's data, treating each control as a candidate collider, for both pooled OLS and country fixed effects specifications.

2 Data

```
dat <- read.delim(
  "candidate_papers/claassen_2020/Support_democracy_ajps_correct.tab",
  stringsAsFactors = FALSE
)
```

The dataset contains 4185 country-year observations for 135 countries over 1987–2017.

3 Replication of Claassen (2020) Table 1

3.1 Pooled OLS (Model 1.1)

```
mod_ols <- feols(
  Libdem_VD ~ Libdem_m1 + Libdem_m2 + SupDem_m1 +
    lnGDP_imp_m1 + GDP_imp_grth_m1 + Libdem_regUN_m1 +
    Pr_Muslim + Res_cp_WDI_di_m1,
```

```
  data = dat, vcov = "iid"
)
```

```
published <- c(
  "(Intercept)" = 0.647, Libdem_m1 = 1.141, Libdem_m2 = -0.163,
  SupDem_m1 = 0.267, lnGDP_imp_m1 = 0.015, GDP_imp_grth_m1 = 0.007,
  Libdem_regUN_m1 = 0.008, Pr_Muslim = -0.002, Res_cp_WDI_di_m1 = -0.367
)
replicated <- round(coef(mod_ols), 3)
comparison <- data.frame(
  Term = names(replicated),
  Published = published[names(replicated)],
  Replicated = as.numeric(replicated)
)
kable(comparison, digits = 3,
      caption = "Pooled OLS: Published vs. Replicated coefficients (N = 2,435)")
```

Table 1: Pooled OLS: Published vs. Replicated coefficients (N = 2,435)

	Term	Published	Replicated
(Intercept)	(Intercept)	0.647	0.659
Libdem_m1	Libdem_m1	1.141	1.141
Libdem_m2	Libdem_m2	-0.163	-0.163
SupDem_m1	SupDem_m1	0.267	0.273
lnGDP_imp_m1	lnGDP_imp_m1	0.015	0.015
GDP_imp_grth_m1	GDP_imp_grth_m1	0.007	0.007
Libdem_regUN_m1	Libdem_regUN_m1	0.008	0.008
Pr_Muslim	Pr_Muslim	-0.002	-0.002
Res_cp_WDI_di_m1	Res_cp_WDI_di_m1	-0.367	-0.373

Coefficients match closely. The small discrepancy in `SupDem_m1` (0.267 vs. 0.273) is due to using the pre-computed lag `SupDem_m1` rather than `plm::lag(SupDem_trim, 1)`, which may handle edge observations differently.

3.2 Dynamic Fixed Effects

```
mod_fe <- feols(
  Libdem_VD ~ Libdem_m1 + Libdem_m2 + SupDem_m1 +
    lnGDP_imp_m1 + GDP_imp_grth_m1 + Libdem_regUN_m1 +
    Res_cp_WDI_di_m1 | Country,
  data = dat, vcov = "iid"
)
```

```
fe_coefs <- data.frame(
  Term = names(coef(mod_fe)),
  Estimate = as.numeric(round(coef(mod_fe), 4)),
  SE = as.numeric(round(sqrt(diag(vcov(mod_fe))), 4))
)
kable(fe_coefs, digits = 4,
      caption = "Dynamic FE estimates (N = 2,435, 135 countries)")
```

Table 2: Dynamic FE estimates (N = 2,435, 135 countries)

Term	Estimate	SE
Libdem_m1	0.9792	0.0207
Libdem_m2	-0.1571	0.0201
SupDem_m1	-0.0158	0.2156
lnGDP_imp_m1	-0.3916	0.2103
GDP_imp_grth_m1	-0.0142	0.0134
Libdem_regUN_m1	-0.0194	0.0280
Res_cp_WDI_di_m1	0.0019	0.4518

With country fixed effects, the coefficient on `SupDem_m1` drops to approximately zero ($\hat{\beta}$ = -0.016, $p = 0.942$), consistent with the Nickell bias problem discussed in Claassen (2020).

4 IVB Decomposition

4.1 Setup

We define:

- **Treatment** (d): `SupDem_m1` (lagged public support for democracy)
- **Outcome** (y): `Libdem_VD` (V-Dem Liberal Democracy index)
- **ADL terms** (always in w): `Libdem_m1`, `Libdem_m2`
- **Candidate colliders** (z): each of the five controls, one at a time

For each candidate collider z, the "short" model excludes z and the "long" model includes it. The IVB identity $\hat{\beta}^* - \hat{\beta} = -\hat{\theta}^* \cdot \hat{\pi}$ is verified numerically.

4.2 Pooled OLS

```
all_controls <- c("lnGDP_imp_m1", "GDP_imp_grth_m1", "Libdem_regUN_m1",
                  "Pr_Muslim", "Res_cp_WDI_di_m1")
ldv_terms <- c("Libdem_m1", "Libdem_m2")

ivb_ols <- lapply(all_controls, function(z_var) {
  w_vars <- c(ldv_terms, setdiff(all_controls, z_var))
  compute_ivb_multi(
    data = dat, y = "Libdem_VD", d_vars = "SupDem_m1",
    z = z_var, w = w_vars, fe = character(), vcov = "iid"
  )
})
names(ivb_ols) <- all_controls

tab_ols <- do.call(rbind, lapply(names(ivb_ols), function(z) {
  r <- ivb_ols[[z]]$results
  data.frame(
    Collider = z,
    beta_short = r$beta_short,
    beta_long = r$beta_long,
    theta = r$theta,
    pi = r$pi,
    IVB = r$ivb_formula,
    check = r$diff_check,
```

```
    stringsAsFactors = FALSE
  )
}))
```

```
tab_ols_print <- tab_ols
tab_ols_print$Collider <- c("Log GDP p.c.", "GDP growth", "Regional democracy",
                            "% Muslim", "Resource dep.")
kable(tab_ols_print, digits = 4, row.names = FALSE,
      caption = "IVB decomposition for each candidate collider (Pooled OLS)",
      col.names = c("Candidate collider $z$", "$\\hat\\beta_{\\text{short}}$",
                    "$\\hat\\beta_{\\text{long}}$", "$\\hat\\theta^*$",
                    "$\\hat\\pi$", "IVB $= -\\hat\\theta^* \\hat\\pi$",
                    "Check"))
```

Table 3: IVB decomposition for each candidate collider (Pooled OLS)

Candidate collider z	$\hat{\beta}_{\text{short}}$	$\hat{\beta}_{\text{long}}$	$\hat{\theta}^*$	$\hat{\pi}$	IVB $= -\hat{\theta}^*\hat{\pi}$	Check
Log GDP p.c.	0.2740	0.2729	0.0148	0.0707	-0.0010	0
GDP growth	0.2739	0.2729	0.0067	0.1383	-0.0009	0
Regional democracy	0.2822	0.2729	0.0084	1.1020	-0.0092	0
% Muslim	0.2719	0.2729	-0.0018	0.5807	0.0010	0
Resource dep.	0.2646	0.2729	-0.3730	0.0223	0.0083	0

The identity holds exactly (check ≈ 0) in all cases. Regional democracy produces the largest IVB (-0.009), reducing the estimated effect of support from 0.282 to 0.273.

4.3 Country Fixed Effects

```
fe_controls <- c("lnGDP_imp_m1", "GDP_imp_grth_m1", "Libdem_regUN_m1",
                 "Res_cp_WDI_di_m1")

ivb_fe <- lapply(fe_controls, function(z_var) {
  w_vars <- c(ldv_terms, setdiff(fe_controls, z_var))
  compute_ivb_multi(
    data = dat, y = "Libdem_VD", d_vars = "SupDem_m1",
    z = z_var, w = w_vars, fe = "Country", vcov = "iid"
  )
})
names(ivb_fe) <- fe_controls

tab_fe <- do.call(rbind, lapply(names(ivb_fe), function(z) {
  r <- ivb_fe[[z]]$results
  data.frame(
    Collider = z,
    beta_short = r$beta_short,
    beta_long = r$beta_long,
    theta = r$theta,
    pi = r$pi,
    IVB = r$ivb_formula,
```

```
    check = r$diff_check,
    stringsAsFactors = FALSE
  )
}))
```

```
tab_fe_print <- tab_fe
tab_fe_print$Collider <- c("Log GDP p.c.", "GDP growth",
                           "Regional democracy", "Resource dep.")
kable(tab_fe_print, digits = 4, row.names = FALSE,
      caption = "IVB decomposition for each candidate collider (Country FE)",
      col.names = c("Candidate collider $z$", "$\\hat\\beta_{\\text{short}}$",
                    "$\\hat\\beta_{\\text{long}}$", "$\\hat\\theta^*$",
                    "$\\hat\\pi$", "IVB $= -\\hat\\theta^* \\hat\\pi$",
                    "Check"))
```

Table 4: IVB decomposition for each candidate collider (Country FE)

Candidate collider z	$\hat{\beta}_{\text{short}}$	$\hat{\beta}_{\text{long}}$	$\hat{\theta}^*$	$\hat{\pi}$	IVB $= -\hat{\theta}^*\hat{\pi}$	Check
Log GDP p.c.	0.0007	-0.0158	-0.3916	-0.0420	-0.0165	0
GDP growth	-0.0281	-0.0158	-0.0142	0.8680	0.0123	0
Regional democracy	-0.0138	-0.0158	-0.0194	-0.1015	-0.0020	0
Resource dep.	-0.0157	-0.0158	0.0019	0.0639	-0.0001	0

With fixed effects, log GDP per capita produces the largest IVB (-0.016), pulling the support coefficient from near-zero positive to negative.

5 Literature Support for Collider Status

The IVB formula diagnoses **how much** each control shifts the treatment estimate, but **whether** a control is a collider depends on substantive knowledge: does the outcome d_{it} (democracy) cause the control z_{it}?

5.1 Democracy → GDP growth

There is strong evidence that democracy causes economic growth:

- **Acemoglu, Naidu, Restrepo & Robinson (2019, JPE):** Democratization increases GDP per capita by ~20% over 25 years. Uses the same ADL panel framework, system GMM, and IV (regional waves of democratization). This is the current consensus result.
- **Papaioannou & Siourounis (2008, EJ):** Democratization associated with ~1 pp/year additional growth using before-after event study designs.
- **Madsen, Raschky & Skali (2015, EER):** Using linguistic-distance-weighted foreign democracy as IV, find large positive effects of democracy on income over 500 years.
- **Doucouliagos & Ulubasoglu (2008, AJPS):** Meta-analysis of 84 studies finds robust indirect effects of democracy on growth via human capital, inflation, and economic freedom.

Implication: GDP growth is a plausible collider. Democracy d_{it} causes GDP growth z_{it}, and public support s_{it} may also affect GDP growth via political stability. Including GDP growth as a control opens a non-causal path.

5.2 Democracy → Regional democracy

There is moderate-to-strong evidence of democratic diffusion:

- **Gleditsch & Ward (2006, IO):** Democratic neighborhood increases probability of democratic transition and decreases probability of reversal. Markov transition model with panel data.
- **Brinks & Coppedge (2006, CPS):** Countries converge toward the mean democracy level of contiguous neighbors. Robust spatial diffusion effects.
- **Leeson & Dean (2009, AJPS):** Democratic "dominoes" fall, but modestly: countries absorb ~11% of changes in neighbors' democracy levels. Spatial econometrics with 150 years of data.
- **Teorell (2010, Cambridge UP):** Regional democratic organizations and democratic neighbors are robust predictors of democratization across 165 countries.

Additionally, regional democracy (`Libdem_regUN_m1`) has a **mechanical** component: country i's democracy level d_{it} is part of the regional average z_{it}. This makes the collider channel virtually certain.

Implication: Regional democracy is a collider both through genuine diffusion and through mechanical composition. This is the control with the largest IVB in the pooled OLS specification.

6 Summary

```r
summary_all <- data.frame(
  Control = c("Log GDP p.c.", "GDP growth", "Regional democracy",
              "% Muslim", "Resource dep."),
  `Y causes Z` = c("Strong", "Strong", "Moderate + mechanical",
                   "No (time-invariant)", "Weak/reverse"),
  `Key reference` = c("Acemoglu et al. 2019", "Acemoglu et al. 2019",
                      "Gleditsch & Ward 2006", "---", "Ross 2001"),
  `IVB (OLS)` = round(tab_ols$IVB, 4),
  `IVB (FE)` = c(round(tab_fe$IVB[tab_fe$Collider == "lnGDP_imp_m1"], 4),
                 round(tab_fe$IVB[tab_fe$Collider == "GDP_imp_grth_m1"], 4),
                 round(tab_fe$IVB[tab_fe$Collider == "Libdem_regUN_m1"], 4),
                 NA,
                 round(tab_fe$IVB[tab_fe$Collider == "Res_cp_WDI_di_m1"], 4)),
  check.names = FALSE, stringsAsFactors = FALSE
)
kable(summary_all, caption = "Summary: Literature evidence and IVB magnitudes")
```

Table 5: Summary: Literature evidence and IVB magnitudes

Control	Y causes Z	Key reference	IVB (OLS)	IVB (FE)
Log GDP p.c.	Strong	Acemoglu et al. 2019	-0.0010	-0.0165
GDP growth	Strong	Acemoglu et al. 2019	-0.0009	0.0123
Regional democracy	Moderate + mechanical	Gleditsch & Ward 2006	-0.0092	-0.0020
% Muslim	No (time-invariant)	—	0.0010	NA
Resource dep.	Weak/reverse	Ross 2001	0.0083	-0.0001

Two of the five controls in Claassen (2020) — GDP-related variables and regional democracy — have well-documented causal links from democracy. The IVB formula quantifies the resulting bias: regional democracy produces the largest bias in pooled OLS (-0.009, ~3% of $\hat{\beta}$), while log GDP per capita produces the largest bias in the FE specification (-0.016). These magnitudes are modest in absolute terms, but they demonstrate the practical applicability of the IVB decomposition to real-world empirical research.

7 References

- Acemoglu, D., Naidu, S., Restrepo, P., & Robinson, J. A. (2019). Democracy Does Cause Growth. *Journal of Political Economy*, 127(1), 47–100.
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